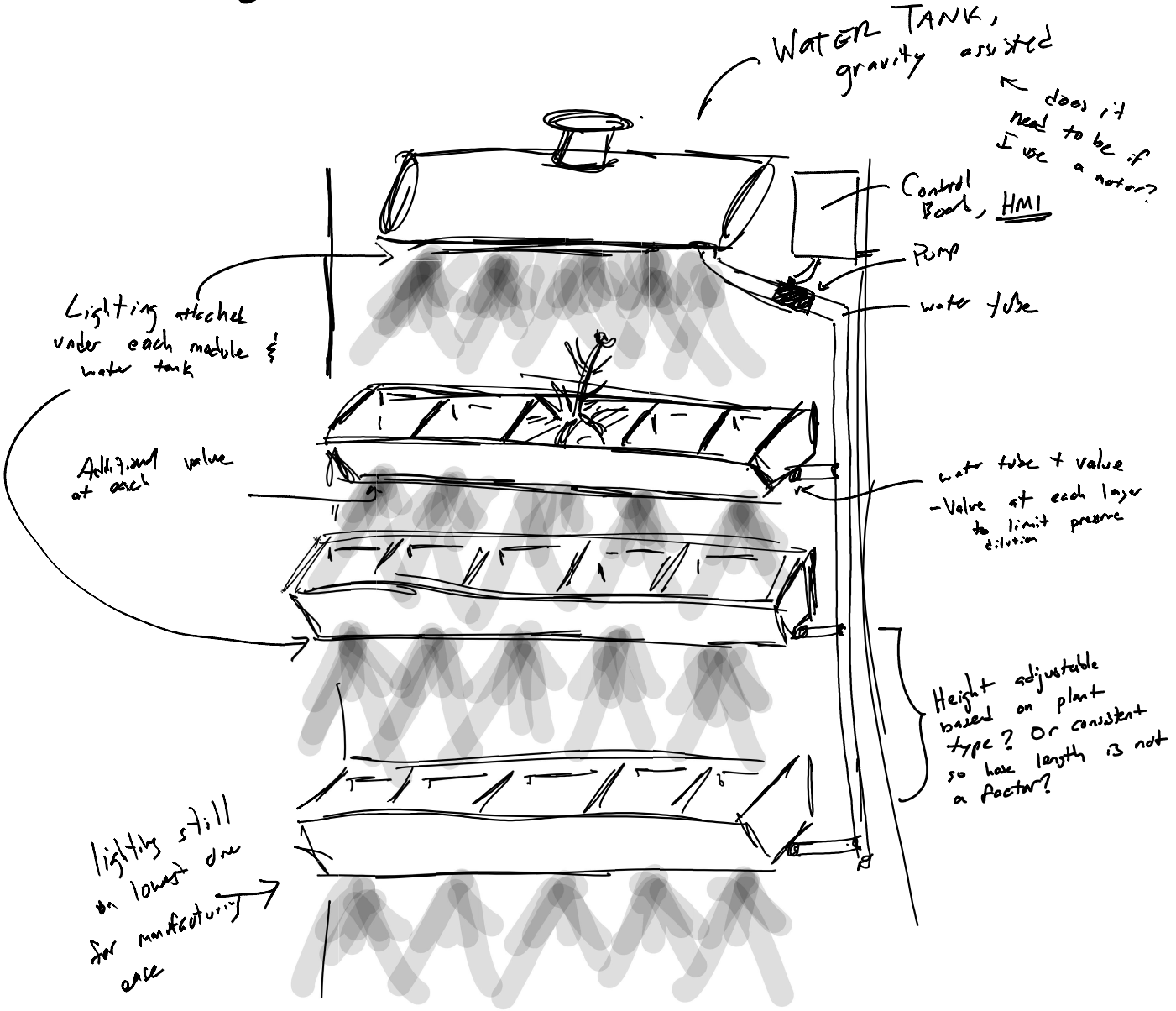
INTRO

I want to create a plant wall that is autonomous because I'm terrible at taking care of them

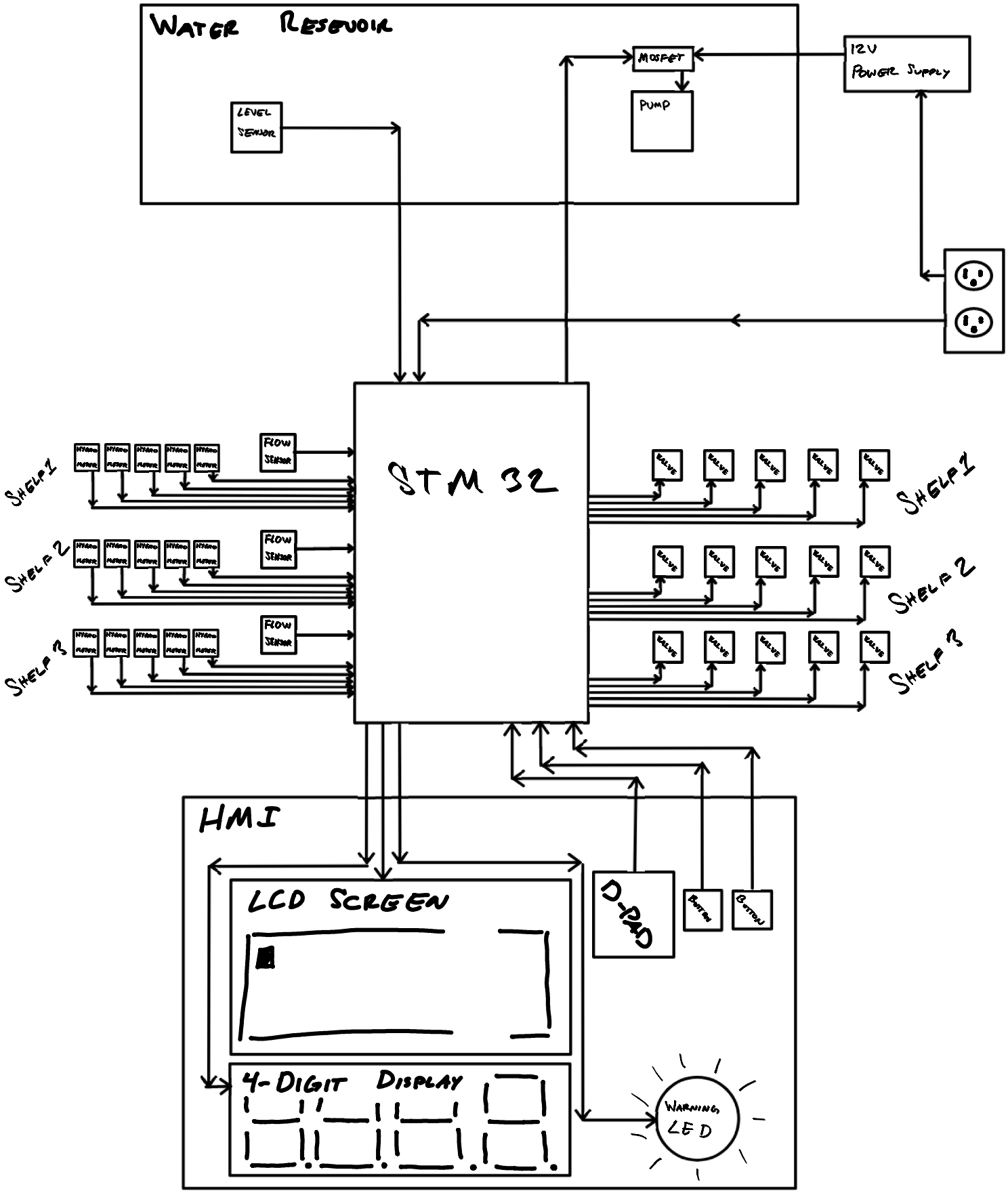
- covering watering & lighting
- Just choose amount of light & water
- Selectable plants to have pre-built profiles?
- Stackable

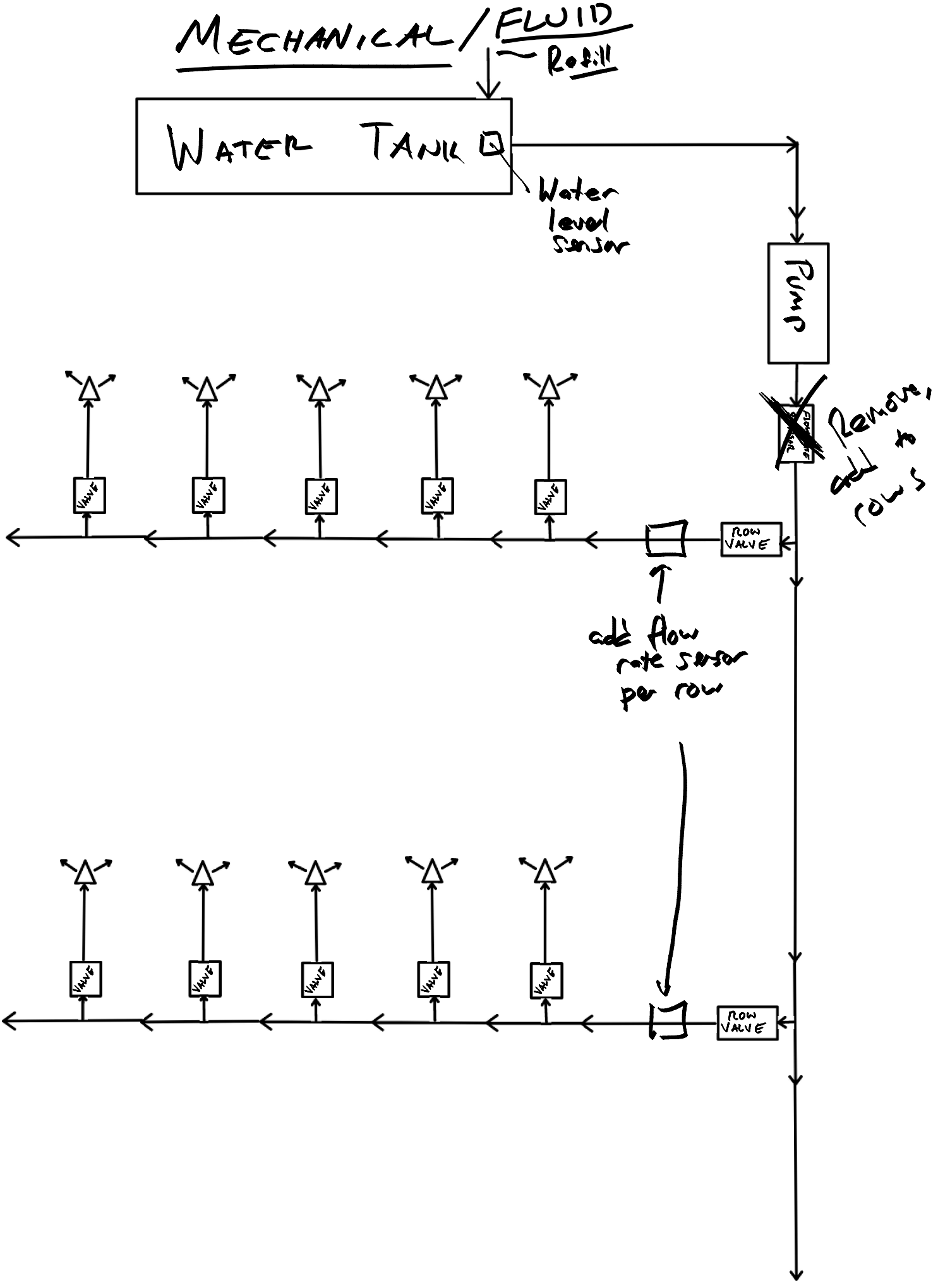


REQUIREMENTS

- Time Limitation: Get it to "Good enough" in 1 month — Month of November 2026
  - Functional electrical system
  - Firmware created
    - Sensor checks
    - plant position checks
  - Mechanical for 2 shelves, 3 plants made
    - Allows for mixing/matching locations to prove out logic
    - Includes plumbing, plant box & row
    - Does not include lighting
  - HMI Functioning to select water amount & time
    - Include functioning buttons, not visible clock.
- Easy for my wife to use (who is not in the SW world), because this is going on her office wall
- In depth understanding of Embedded systems: I do SW work & Testing, but have a gap in knowledge of the Hardware/Firmware part. Want to do the whole process to understand.
- Using RUST to write it — smaller/quicker than C, so want to learn as well
- System to understand which plants exist, to have safety for water values
- Normally closed values, so lower power draw & low leak risk if power loss

System Diagram




MECHANICAL / FLUID
~ Robin
WATER TANK
Water level sensor
PUMP
Remove, add to rows
VALVE
FLOW VALVE
add flow rate sensor per row

HARDWARE

- **Pump / Motor**: Move the water, auto windshield washer pump should be reliable enough, common enough
- **Valves**: For each module, and each row. Normally Closed!!!
- **tubing**: Auto windshield washer diameter, makes it consistent/cheap
- **Plant holder**: Module to hold plants
- **Plant Row**: Row of plants, undefined width yet, possibly varying widths
- **WATER TANK**: Holds the water. Needs to be easy to fill, and hold enough for a week or 2
- **STM32**: To Control the whole thing
- **LCD Screen**: To display info, set profiles
- **D-Pad**: Control HMI to enter Profiles
- **Button(s)**: ENTER for menu, Circle button to enter settings
- **Flow Rate Sensor**: EACH Row instead?
- **LED**: For lights to Feed from
- **Water Level Sensor**: To not run out of water or empty tubes
- **Small LED**: HMI Warning -> Low light, fault, etc.
- **4 DIGIT, 7 FIGURE DISPLAY**: I have one, so I want to use it for the clock

Firmware & Embedded SW

- Checks for # of connected sensors
- Checks for Sensor readings to ensure in spec
  - rate
  - range
- Use hygrometer sensor as presence sensor
- make sure things like flowrate can be updated without a flash needed, in EEPROM

Task Cycles:

- 20 Hz? 100 Hz?
  - depends on flow sensor & physical rate of flow.

- Check for # of rows/plants, allow that many spots in the HMI, hide the rest
- Check fluid level sensor for warning LED, add HMI warning

ELECTRICAL

- Communication protocol:
  - Want adding sensors (from more rows) to be easy
  - Comm rate? flow rate, 100 hz? How fast can I get?
  - I^2C seems like a good option
    - relatively simple
    - Minimal wiring, easy to add new rows
    - Never used, good to learn
    - Standard is 100 kHz, more than fast enough

- Inputs/outputs
  - Max: - 1x overall system
    - 1x water level sensor
    - 1x Pump
  - 5x Rows
    - 1x flow sensor / row
  - 10x plants / row
    - 1x value / plant
    - 1x LED / plant
  - max = 102 possible I/O
  - STM32 nucleo is: 114
    - 5V though, need 12 for pump

TUNING/MODELING

- With flow meter, will need to tune the actual water out vs the calculated.

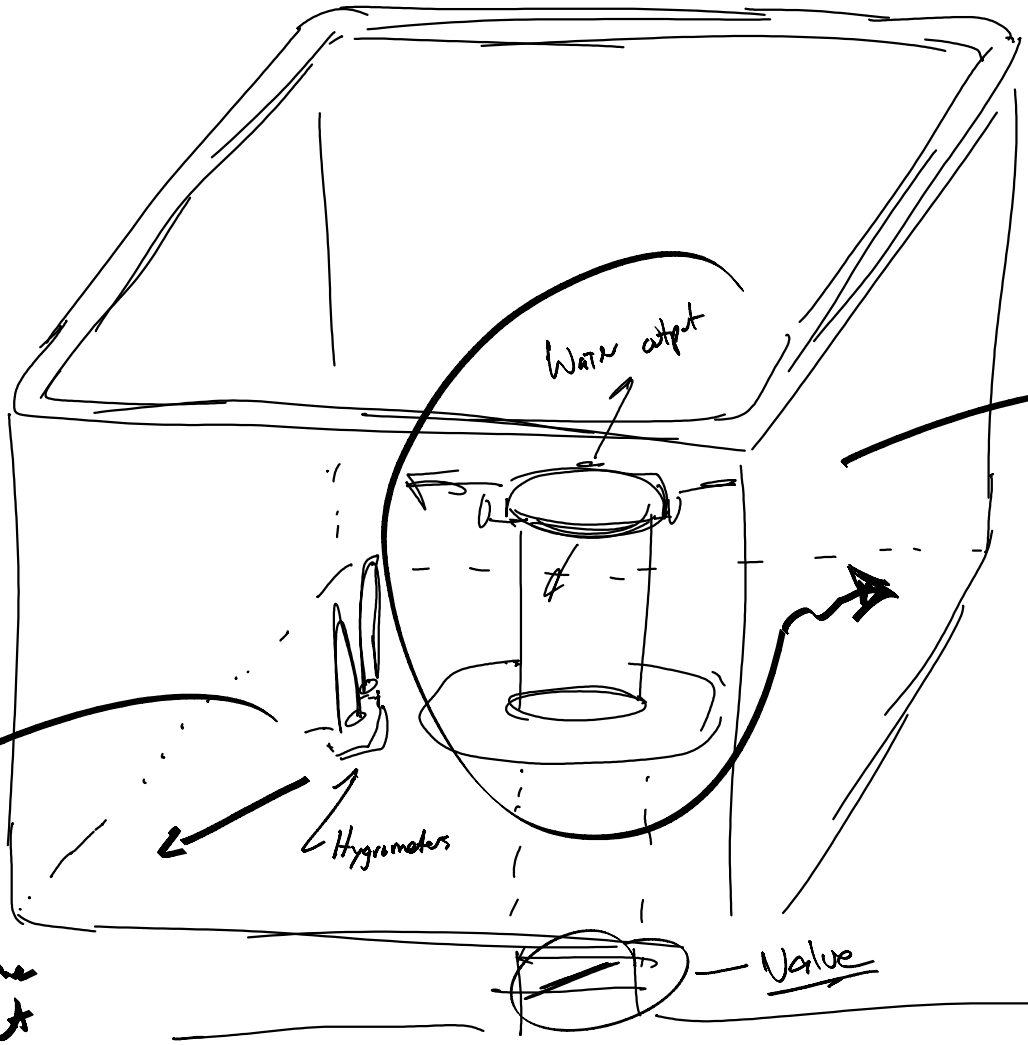
  3 sections:
  - valve opening
  - valve steady state open
  - valve closing

  - Probably affected by the total # of shelves, what is the min # of sensors I can get away w/?

- Hygrometer: water amount vs. soil moisture level
  - probably not linear, will need a curve parameter

PLANT HOLDER

INITIAL IDEA



Move to corner, but up until soil line to not interfere w/ plant growth

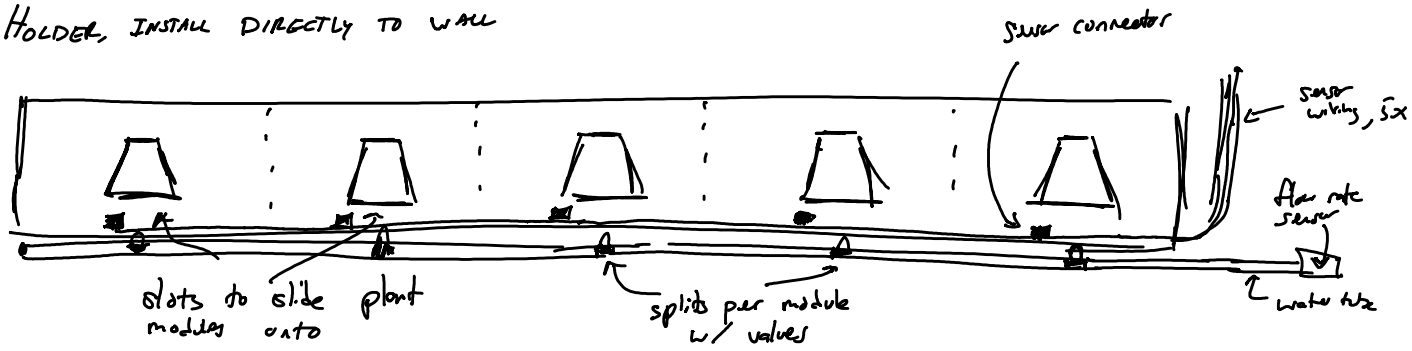
Move to opposite side if wise, will add time to read, but will be more accurate to the total soil moisture level.

- Maybe add reading delay

- Drainage
- Quick connect/disconnect
- Water + Sensor into id
- Add flow meter here? or only per row? per whole system?

PLANT ROW

HOLDER, INSTALL DIRECTLY TO WALL

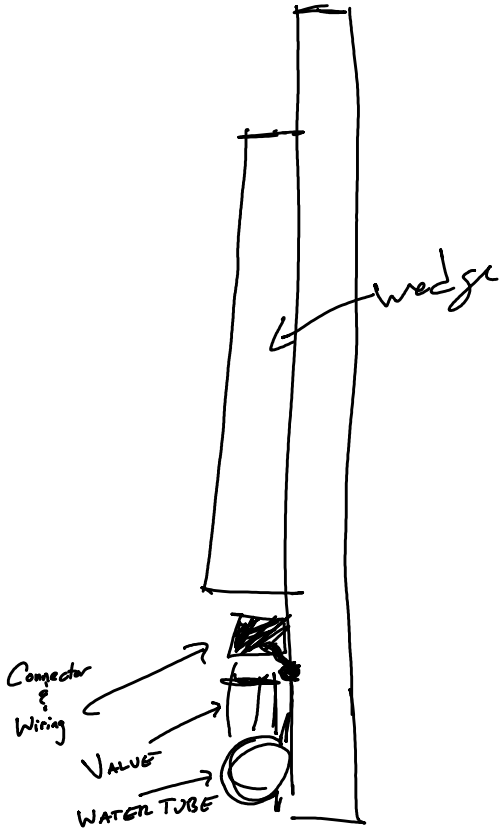


- There should be a limit switch added per slot to allow for the valves to work or not.

- DRAINAGE:
  - If tuned well, drainage should be minimal
  - Small cubby built into each module to dump manually?
  - don't want extra plumbing unless necessary.

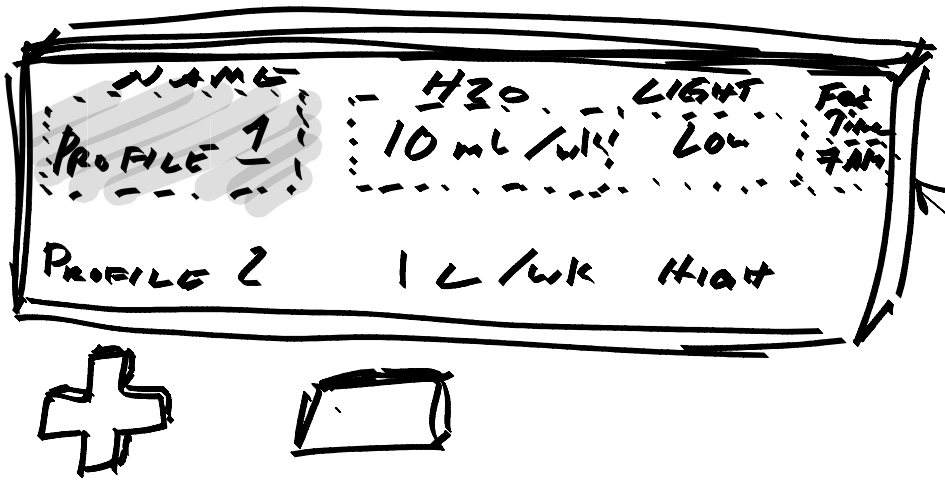
- 5x plants per row,
  sized for max 1.5' plant height.

SIDE VIEW



HMI

- Allow the user to create custom profiles
- Pre-made profiles in drop-down for various plants



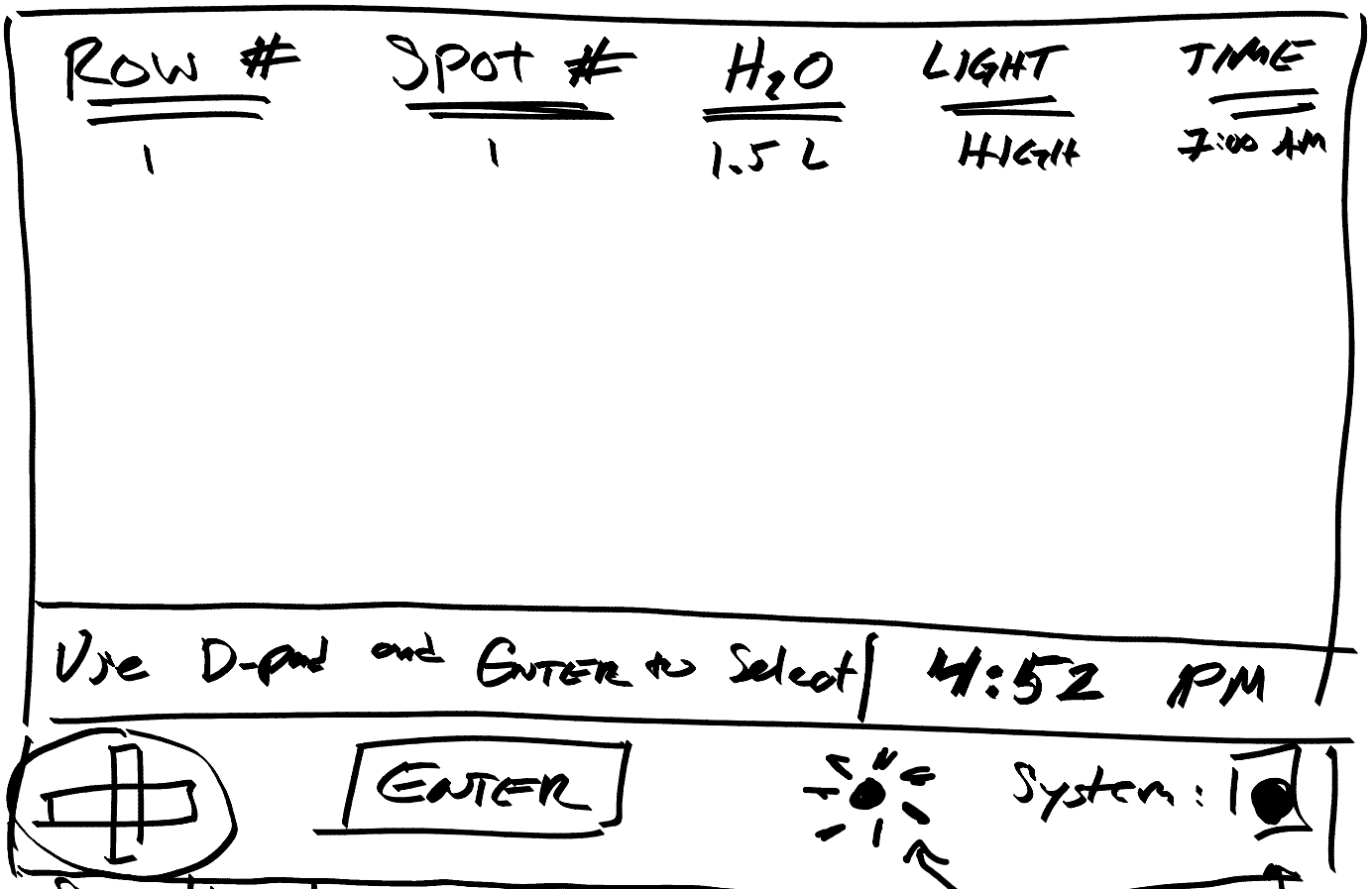
- Simple LED
- non-touch

D-pad for interface

ENTER BUTTON

- Should include clock & watering time

HMI AGAIN



- Scroll down
- Scroll across for more options?

Button

FAULT/WARNING LED